

An Analytic Origin for the Provenance Bias in Radial Migration: A Vertical Form Factor, Corotation Trapping, and the Spiral J_R - h_Z Relation

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ABSTRACT

Radial migration in disk galaxies acts preferentially on vertically cold stars – the “provenance bias” seen in N -body simulations (C. Vera-Ciro et al. 2016; D. Mikkola et al. 2020) and so far understood only qualitatively. This paper derives it from first principles: a star’s coupling to a spiral is reduced, on hotter vertical orbits, by the orbit-average of the potential’s $e^{-k|z|}$ fall-off above the plane – a vertical form factor F that proves to be the exact vertical coupling of the spiral at corotation, so vertically cold stars are trapped and migrate preferentially. The derived bias is regime-dependent: single-resonance trapping floors it near ~ -7 percent, whereas resonance overlap drives the disk into a diffusive regime with a predicted ~ -20 to -27 percent band. By the Chirikov criterion the Milky Way bar and spiral corotation resonances sit at this overlap threshold for physical spiral strengths, with cold stars crossing it first, so the diffusive channel is itself cold-selective – matching the strong cold-migrator preference of the simulations. The same form factor unifies the bias with the empirical spiral J_R^{max} - h_Z relation (P. A. Palicio et al. 2025), as both depend on disk thickness only through $\alpha \equiv kh_Z$: the predicted slope reproduces the measured $3.7 \times 10^{-2} L_\odot/\text{kpc}$ and, being set by the radial-to-vertical heating anisotropy, should flatten with stellar age – a clean, falsifiable test for mono-age samples in Gaia-APOGEE-LAMOST.

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1. INTRODUCTION

Transient, recurrent spiral arms (J. A. Sellwood & R. G. Carlberg 1984; E. D’Onghia et al. 2013; J. Baba et al. 2013; J. A. Sellwood & R. G. Carlberg 2014) churn stellar angular momentum at the corotation resonance, moving stars radially with little heating (J. A. Sellwood & J. J. Binney 2002; R. Roškar et al. 2008). A robust but only qualitatively understood feature of this process is its *vertical selectivity*: in N -body disks the stars that migrate are preferentially those on nearly circular orbits confined to the plane. C. Vera-Ciro et al. (2016) named this the “provenance bias” and showed that it – rather than vertical heating – sets the vertical structure of migrators: the strongest migrators are preferentially the kinematically cool stars, those with vertical velocity dispersions below the local mean (their Figure 4). D. Mikkola et al. (2020) confirmed that the importance of migration at large vertical action depends on spiral strength, and explicitly flagged an analytic description as outstanding.

Independently, P. A. Palicio et al. (2025) report a linear relation, found across simulated spiral galaxies and

satisfied by the Milky Way, between the maximum radial action of spiral-arm stars and the disk vertical scale h_Z – measured there as an exponential vertical scale-length – $J_R^{\text{max}} = a h_Z + b$ with $a \approx 3.7 \times 10^{-2} L_\odot/\text{kpc}$ (see also Y. Jia et al. 2024); the slope is not derived from first principles.

This paper provides the missing dynamics. K. J. Daniel & R. F. G. Wyse (2015, 2018) derived an analytic corotation-capture criterion in which the maximum random energy for capture scales with the spiral amplitude $|\Phi_s|$, so that low-dispersion stars migrate preferentially – but their criterion is strictly in-plane. This paper extends the problem to three dimensions and shows that the vertical structure enters through a single object, the vertical form factor $F(J_z, k)$ (Section 2), identified as the exact vertical Fourier coefficient of the spiral in the corotation-resonant Hamiltonian (Section 3). The resulting pendulum traps vertically cold stars preferentially; single-resonance trapping is too weak to explain the strong cold-migrator preference of the simulations, and Section 4 shows that the Milky Way sits at the B. V. Chirikov (1979) resonance-overlap threshold, placing the

churning in the diffusive regime. Section 5 then recovers the provenance-bias magnitude and the $J_R^{\max}$ – h_z slope and its age dependence, and Section 6 discusses the synthesis.

2. THE VERTICAL FORM FACTOR

A razor-thin spiral perturbation of radial wavenumber k has a potential that satisfies Laplace’s equation above the disk and therefore decays as $e^{-k|z|}$ (J. Binney & S. Tremaine 2008). A star executing a vertical oscillation $z(\theta_z; J_z)$ samples the spiral through the angle-average of that decay,

$$F(J_z, k) = \langle e^{-k|z(\theta_z, J_z)|} \rangle_{\theta_z} = \frac{2}{\pi} \int_0^{z_m} dz e^{-kz} \frac{\nu_z}{|v_z|}, \quad (1)$$

with z_m the turning point, ν_z the vertical frequency, E_z the conserved vertical-oscillation energy and $v_z = \sqrt{2[E_z - \Phi(z)]}$. Equation (1) is evaluated on the self-consistent isothermal sheet (L. Spitzer 1942), $\Phi(z) = 2 \ln \cosh z$ in units $G = \sigma_z = z_0 = 1$, for which the only parameter is the dimensionless thickness $\alpha \equiv kz_0$. By construction $F(0, k) = 1$ – a mid-plane orbit feels the full in-plane amplitude – and F decreases monotonically with vertical action (Figure 1). Physically, F is the average amplitude a star feels – the time its orbit spends near the plane, weighted by the spiral’s $e^{-k|z|}$ reach. It makes precise the “spend more time near the mid-plane” half of the verbal mechanism; that this orbit-average is itself the coupling strength entering the corotation resonance is shown in Section 3.

For the Milky Way a radial wavelength $\lambda_R \sim 3 \text{ kpc}$ ($k = 2\pi/\lambda_R \sim 2 \text{ kpc}^{-1}$; M. J. Reid et al. 2019) and a vertical scale $z_0 \sim 0.4 \text{ kpc}$ (J. Bland-Hawthorn & O. Gerhard 2016) give $\alpha \sim 0.84$ – of order unity, so the vertical coupling is appreciably selective rather than negligible; the resulting bias is quantified in the trapping and overlap analyses below.

Finite-thickness correction.—The $e^{-k|z|}$ kernel is the razor-thin leading term; the arm overdensity is in fact disk material of finite vertical extent h_s (comparable to the disk scale z_0), and convolving the thin-sheet kernel with that profile rounds the mid-plane cusp (A. Toomre 1964). For an exponential arm profile $g(z) \propto e^{-|z|/h_s}$, with h_s the arm’s vertical scale height, the softened kernel is $K(z) = [\beta e^{-k|z|} - k e^{-\beta|z|}]/(\beta - k)$ with $\beta = 1/h_s$ – the thin-sheet kernel convolved with g and normalized to $K(0) = 1$, a direct solution of Poisson’s equation (J. Binney & S. Tremaine 2008). Its orbit average is

$$F_{\text{soft}}(J_z; \alpha, h) = \frac{F(J_z, \alpha) - \alpha h F(J_z, 1/h)}{1 - \alpha h}, \quad h \equiv h_s/z_0, \quad (2)$$

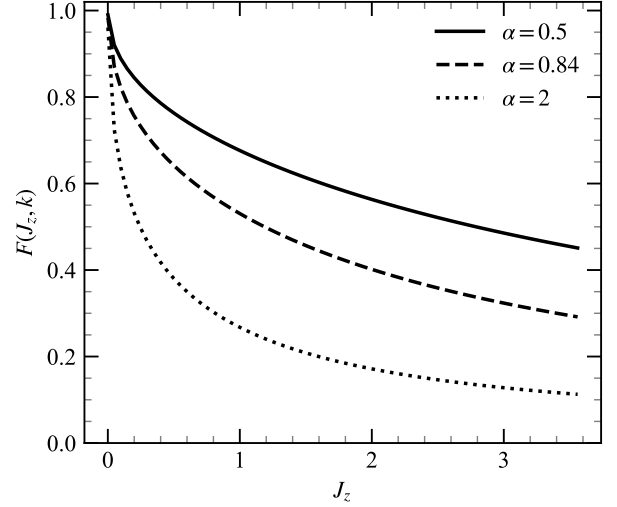


Figure 1. The vertical form factor $F(J_z, k) = \langle e^{-k|z|} \rangle$ (Equation (1)) versus vertical action, for three wavenumbers $\alpha = kz_0$ ($\alpha = 0.84$ approximates the Milky Way). F is the orbit-averaged spiral amplitude a star feels: it equals unity for a mid-plane orbit by construction and falls monotonically with J_z , since a vertically hot star spends more of its orbit far from the plane where the arm is weak, and the suppression steepens for more tightly wound (larger- α) arms. It makes “cold stars couple more readily” quantitative – the whole mechanism in a single object, to which every later figure traces back.

a combination of the bare form factor at the two wavenumbers (derived in Appendix A). It recovers F as $h \rightarrow 0$ and tends to unity as $h \rightarrow \infty$. F_{soft} replaces the razor-thin F when evaluating the Milky-Way provenance bias in Section 3.

3. THE COROTATION-RESONANT HAMILTONIAN AND THE TRAPPED WEIGHT

This section shows that F is not an inserted weight but the vertical structure factor of corotation capture. Write the background in action-angle variables $(J_R, L_z, J_z; \theta_R, \theta_\phi, \theta_z)$ with frequencies (κ, Ω, ν_z) . A steady m -armed, tightly wound spiral of pattern speed Ω_p is

$$\Phi_s = \text{Re} \left[\Phi_0(R) e^{-k|z|} e^{i(m\phi + \int dR k - m\Omega_p t)} \right], \quad (3)$$

whose Fourier expansion in the background angles,

$$\Phi_s = \sum_{n_R, n_z} \Phi_{n_R n_z}(J_R, L_z, J_z) e^{i(n_R \theta_R + m \theta_\phi + n_z \theta_z - m \Omega_p t)}, \quad (4)$$

has coefficients that factorize into independent radial and vertical fast-angle averages; here n_R and n_z are the integer harmonics of the radial and vertical angles (the

azimuthal harmonic is fixed at the arm number m). The (n_R, n_z) term is resonant when its phase is stationary, $n_R \kappa + n_z \nu_z + m(\Omega - \Omega_p) = 0$. At corotation, $\Omega(L_z) = \Omega_p$, this is the $n_R = n_z = 0$ harmonic – the part of the spiral averaged over both the radial epicycle and the vertical orbit – so the resonant coupling is

$$\begin{aligned} \Psi(J_R, J_z) &= \langle \Phi_s \rangle_{\theta_R, \theta_z} \\ &= \Phi_0(R_{\text{CR}}) J_0(ka_R) F(J_z, k), \end{aligned} \quad (5)$$

the spiral averaged over the fast angles θ_R, θ_z and evaluated at the corotation radius R_{CR} (where $\Omega = \Omega_p$), with epicyclic amplitude $a_R = \sqrt{2J_R/\kappa}$. The coupling factorizes into the classical radial reduction factor $G(J_R) \equiv J_0(ka_R)$ (the Kalnajs/Bessel factor; [J. Binney & S. Tremaine 2008](#)) and the vertical form factor F . This coupling Ψ is the amplitude of the corotation resonance – the depth of the pendulum well that traps migrating stars (Equations (6)–(7) below) – so its vertical factor F directly sets how strongly a star of given J_z is captured and migrates. *The provenance bias is the F -factor of this coupling; the [K. J. Daniel & R. F. G. Wyse \(2015\)](#) radial selectivity is the G -factor.* Equation (5) is the central identification of this paper: $F(J_z, k)$ is the exact vertical Fourier coefficient of the spiral at corotation.

Transforming to the rotating frame (the Jacobi integral $H_J = H_0 - \Omega_p L_z + \Phi_s$ is conserved) and averaging over the fast angles θ_R, θ_z – for which J_R, J_z are adiabatic invariants – leaves the slow Hamiltonian. Expanding about corotation in $p \equiv L_z - L_{\text{CR}}$, where $dH_0/dL_z = \Omega - \Omega_p$ vanishes, gives a pendulum

$$\bar{H}(p, \theta_r) = \frac{1}{2} g p^2 + \Psi \cos \theta_r, \quad g \equiv \left. \frac{d\Omega}{dL_z} \right|_{\text{CR}}, \quad (6)$$

following the action-angle resonant-trapping construction of [G. Monari et al. \(2017\)](#), here extended to carry the vertical degree of freedom through Ψ . The libration (trapped, migrating) island has half-width in p

$$\Delta p = 2\sqrt{\Psi/|g|} \propto \sqrt{F(J_z, k)}, \quad (7)$$

so vertically cold stars (large F) sit in wider libration islands and are trapped preferentially. Counting the trapped phase-space area gives a single-resonance migration weight

$$W(J_z) \propto F(J_z, k)^{1/2}, \quad (8)$$

the island-area-counting law. The *exact* trapped fraction follows from integrating a Gaussian parent distribution of the slow action p , of dispersion σ_p , over the separatrix (Appendix B):

$$W(J_z) = \frac{1}{\pi} \int_0^\pi du \operatorname{erf} \left(\frac{\kappa \sqrt{F(J_z, k)}}{\sqrt{2}} \sin u \right), \quad (9)$$

with a single dimensionless island-width parameter $\kappa \equiv p_{\text{max}}(F=1)/\sigma_p \propto \sqrt{|\Phi_s|}$. Equation (9) reduces to the $\sqrt{F}$ law (8) for narrow islands ($\kappa \rightarrow 0$) and saturates to unity for wide ones (Figure 2). The provenance bias follows by reweighting the equilibrium vertical distribution $dN/dE_z \propto e^{-E_z}/\nu_z$ by the trapped weight $W(J_z)$ and forming the migrator-to-parent dispersion ratio $\sigma_z^{\text{mig}}/\sigma_z^{\text{all}}$; Figure 3 plots the resulting log dispersion ratio $\ln \sigma_z^{\text{mig}}/\sigma_z^{\text{all}}$ (negative; more negative is stronger) at the Milky-Way thickness against the island-width κ . It is *most negative* in the dilute limit – where $W \propto \sqrt{F}$ gives the steepest vertical weighting – and weakens (toward zero) monotonically with κ : the exact trapped census therefore *floors* the bias at the $\sqrt{F}$ value, ≈ -7 percent at the Milky-Way thickness, for any spiral strength. Finite spiral thickness softens this floor further: replacing F by the finite-thickness F_{soft} (2) weakens the ≈ -7 percent floor to ≈ -4 to -5.5 percent across the plausible Milky-Way spiral thickness $h \sim 0.5$ – 1 (Figure 4), a 20–45 percent reduction in magnitude. Even the steeper [K. J. Daniel & R. F. G. Wyse \(2018\)](#) capture-fraction accounting, $W \propto F$, reaches only ≈ -14 percent. Neither single-resonance estimate produces the strong cold-migrator preference seen in the simulations, in which the migrators are drawn overwhelmingly from the kinematically cool population ([C. Vera-Ciro et al. 2016](#)); that preference points to the resonance-overlap regime of Section 4, where W steepens toward the diffusive F^2 limit. The effective weight $W = \exp[-s(1/F - 1)]$ derived for the bias in Section 5, Equation (14), is accordingly a *diffusive* effective model, not the single-resonance census.

4. RESONANCE OVERLAP AND THE DIFFUSIVE REGIME

The single-resonance estimates of Section 3 – the exact trapped census ($\sqrt{F}$, ≈ -7 percent; Figure 3) and the [K. J. Daniel & R. F. G. Wyse \(2018\)](#) capture fraction ($\propto F$, ≈ -14 percent) – are both too weak to produce the strong cold-migrator preference of the simulations. Single isolated corotation trapping is therefore not the operative regime. Whether the Milky Way is instead in the resonance-overlap (stochastic, diffusive) regime is tested with the [B. V. Chirikov \(1979\)](#) criterion.

For a flat rotation curve ($\Omega = V_c/R$, $|g| = 1/R^2$) the corotation libration half-width (7) is a fixed fraction of the corotation radius,

$$\Delta R/R = 2\sqrt{\epsilon G F}, \quad \epsilon \equiv |\Phi_s|/V_c^2, \quad (10)$$

and two patterns with corotation radii $R_{\text{in}} < R_{\text{out}}$ overlap when the sum of their island half-widths exceeds

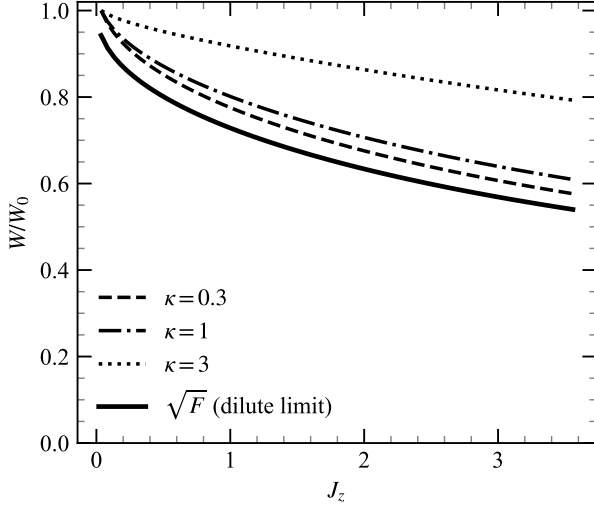


Figure 2. The exact corotation trapped weight (9) (normalized to its mid-plane value) versus vertical action, for three island-width parameters κ . The trapped fraction is steepest in the dilute, narrow-island limit ($\kappa \rightarrow 0$), where it follows $\sqrt{F}$ (dashed), and flattens toward no selectivity as the island widens. Since the corotation coupling is $\propto F$, this weight inherits the form factor’s vertical-action dependence: F decides who migrates, picking out cold stars preferentially.

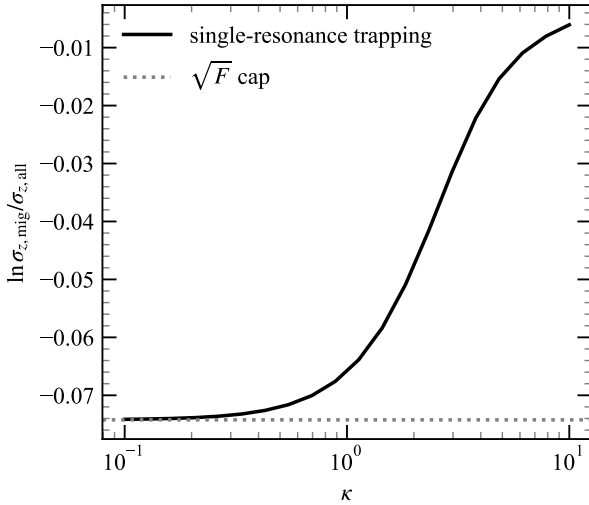


Figure 3. The single-resonance provenance bias $\ln \sigma_{z,\text{mig}}/\sigma_{z,\text{all}}$ at the Milky-Way thickness, from the exact trapped fraction (9), versus the island-width parameter κ . It is *bounded*: most negative in the dilute $\sqrt{F}$ limit (≈ -7 percent, dotted) and weaker for wider islands. One resonance is therefore too weak – it cannot exceed ≈ -7 percent for any spiral strength, far short of the cold-migrator preference seen in simulations, which forces the analysis beyond the single-resonance picture.

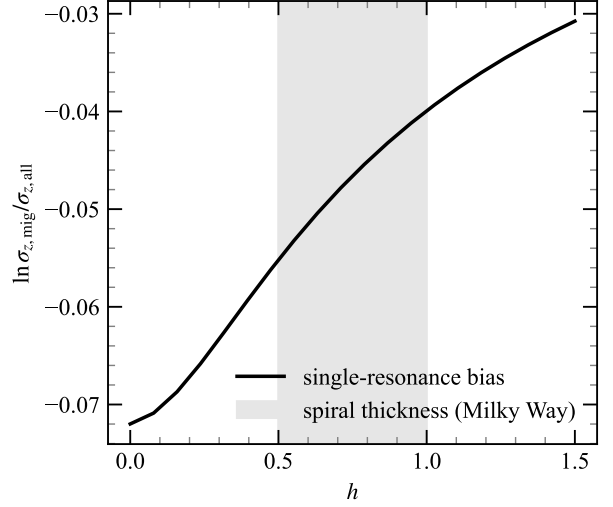


Figure 4. The single-resonance bias at the Milky-Way thickness ($\alpha = 0.84$), computed with the finite-thickness form factor F_{soft} (2) in place of the razor-thin F in the trapped census (9), versus the spiral vertical thickness $h = h_s/z_0$. The razor-thin $\sqrt{F}$ floor (≈ -7 percent at $h = 0$) weakens to ≈ -4 to -5.5 percent across the plausible arm thickness $h \sim 0.5\text{--}1$ (shaded). Finite arm thickness only widens the gap: a single resonance falls well short of the observed bias.

their separation,

$$S \equiv \frac{\Delta R_{\text{in}} + \Delta R_{\text{out}}}{R_{\text{out}} - R_{\text{in}}} = 2\sqrt{\epsilon G F} \frac{\Omega_{\text{in}} + \Omega_{\text{out}}}{\Omega_{\text{in}} - \Omega_{\text{out}}} \geq 1. \quad (11)$$

For the Milky Way bar ($\Omega_p \approx 38 \text{ km s}^{-1} \text{ kpc}^{-1}$, $R_{\text{CR}} \approx 6.0 \text{ kpc}$) and spiral ($\Omega_p \approx 20 \text{ km s}^{-1} \text{ kpc}^{-1}$, $R_{\text{CR}} \approx 11.5 \text{ kpc}$), Equation (11) gives $S = 0.64, 0.91, 1.12$ at $\epsilon = 0.01, 0.02, 0.03$ – so at the physical spiral strength ($|\Phi_s| \sim 1\text{--}3\% V_c^2$) the bar and spiral corotation resonances sit at the overlap threshold $S \sim 1$, the mechanism of I. Minchev & B. Famaey (2010).

Crucially, since $\Delta R \propto \sqrt{F}$, the overlap parameter scales as $S \propto \sqrt{F}$: vertically cold stars cross the threshold first (Figure 5). At the fiducial parameters the cold ($F = 1$) population overlaps at $\epsilon \approx 0.024$, inside the physical band, while the hot ($F = 0.5$) population stays in regular libration until $\epsilon \approx 0.05$. The diffusive churning channel is therefore itself biased toward vertically cold stars, steepening the provenance bias beyond the single-resonance $F^{1/2}$ law toward the F^2 diffusive limit. The Milky Way spiral sector lives at the crossover between regular trapping and overlap diffusion, and that crossover is vertical-action selective.

The crossover weight. —The two regimes are limits of one migration weight. Quasilinear churning diffusion gives an angular-momentum diffusion coefficient $D \propto |\Psi|^2 \propto$

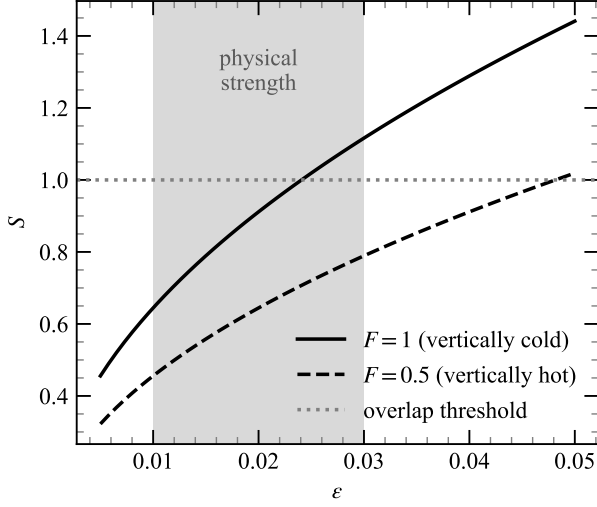


Figure 5. The Chirikov overlap parameter S of the Milky-Way bar and spiral corotation resonances (11), versus spiral strength $\epsilon = |\Phi_s|/V_c^2$, for vertically cold ($F = 1$) and hot ($F = 0.5$) stars. Overlap – the onset of diffusive churning – sets in at $S = 1$, reached within the physical strength band (shaded); and because $S \propto \sqrt{F}$, cold stars cross first, so in that band only the cold population is diffusive while hot stars stay trapped. Overlapping resonances are the resolution to the single-resonance shortfall, and the transition is itself cold-selective.

F^2 , so the diffusive weight is $W \propto F^2$ where resonances overlap, while the trapped census gives $W \propto F^{1/2}$ where they do not. Modeling the weight as a power of the form factor whose exponent runs from the trapping value $\frac{1}{2}$ to the diffusive value 2 through a gate centered on the threshold $S = 1$,

$$W(J_z) = F^p, \quad p = \frac{1}{2} + \frac{3}{2} \frac{S^2}{1 + S^2}, \quad S(J_z) = S_0 \sqrt{F(J_z, k)}, \quad (12)$$

with the single physical parameter S_0 the effective overlap, the predicted bias deepens monotonically from the $\sqrt{F}$ value (≈ -7 percent) to the F^2 value (≈ -27 percent) as S_0 grows (Figure 6). The bar and spiral alone ($S_0 \approx 1$) give ≈ -11 percent; the diffusive band (≈ -20 to -27 percent) is reached at $S_0 \approx 2-3$, the extra overlap expected once the Milky Way’s recurrent transient spirals are included. The crossover sits at the critical form factor $F_{\text{crit}} = 1/S_0^2$ – a critical vertical action separating diffusive (cold) from confined (hot) migrators, a structural prediction. The endpoints are physical; the gate is the minimal model of the crossover sharpness.

First-principles diffusive weight.—The diffusive endpoint is not assumed but follows from kinetic theory. The inhomogeneous Balescu–Lenard corotation

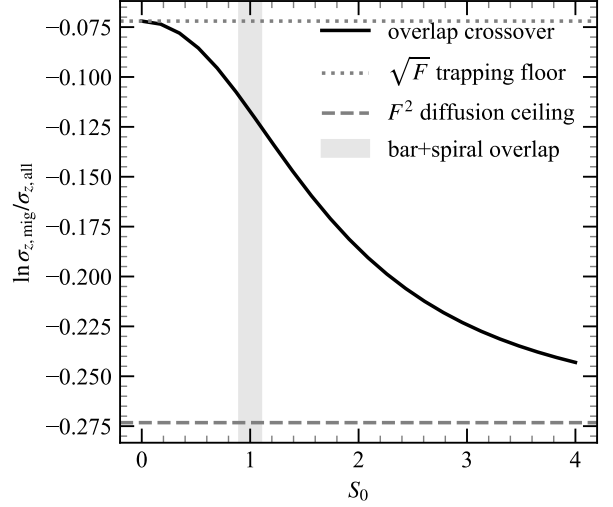


Figure 6. The provenance bias $\ln \sigma_z^{\text{mig}}/\sigma_z^{\text{all}}$ at the Milky-Way thickness from the crossover weight (12), versus the effective resonance overlap S_0 . As S_0 grows the weight steepens from $\sqrt{F}$ (trapping) toward F^2 (diffusion), and the bias deepens from the ≈ -7 percent floor to the ≈ -27 percent F^2 value; the bar-plus-spiral overlap alone (shaded, $S_0 \approx 1$) already gives ≈ -11 percent. Overlap deepens the bias well past the single-resonance floor, further still as the Milky Way’s recurrent transient arms raise S_0 .

diffusion coefficient in angular momentum, $D_{L_z}(J) = \pi \sum_m |\Psi_m(J)|^2 C_m(\Omega)$ (J. Heyvaerts 2010; J.-B. Fouvry et al. 2015), inherits the $|\Psi_m|^2 \propto F^2$ of the resonant coupling (5), so the diffusive migration weight is $W \propto F^2$. A finite libration width $\Delta\omega \propto \sqrt{\Psi} \propto \sqrt{F}$ broadens the resonant denominator (T. H. Dupree 1966); for a Lorentzian spiral spectrum of width W_s this yields the closed form

$$W(J_z) = \frac{F^2}{1 + b\sqrt{F}}, \quad b = \frac{\Delta\omega}{W_s}, \quad (13)$$

interpolating the decorrelated quasilinear limit ($W \propto F^2$, $b \rightarrow 0$) and the coherent, resonance-broadened limit ($W \propto F^{3/2}$, $b \rightarrow \infty$). The provenance bias in this diffusive regime is therefore confined to a narrow *derived* band, ≈ -20 to -27 percent at the Galactic thickness, almost independent of b (Figure 7): the observed bias follows from the disk being in the diffusive (overlap) regime, with no tuned parameter. The crossover weight (12) thus bridges the derived $\sqrt{F}$ value (≈ -7 percent) and this derived diffusive band.

Test-particle validation.—The mechanism is confirmed by direct orbit integration, with no analytic input. Integrating ~ 3000 test particles in MWPotential2014 (J. Bovy 2015) plus an imposed transient spiral that falls off vertically over a scale height H , and weighting each

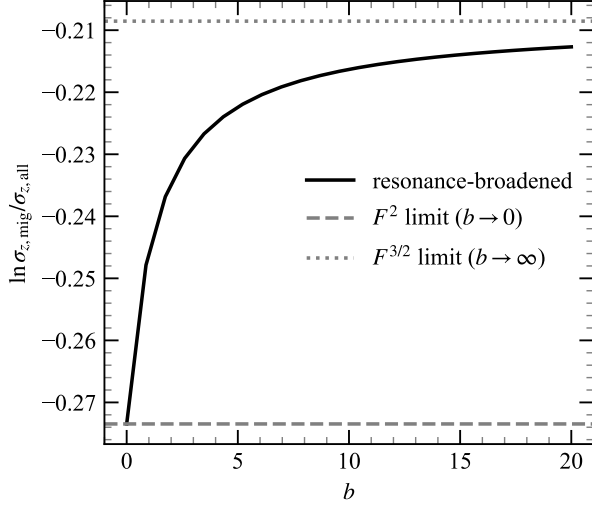


Figure 7. The provenance bias $\ln \sigma_z^{\text{mig}} / \sigma_z^{\text{all}}$ at the Milky-Way thickness from the first-principles Balescu-Leonard diffusive weight (13), versus the resonance-broadening parameter b . It is pinned to the narrow band between its F^2 and $F^{3/2}$ limits (shaded), ≈ -20 to -27 percent, nearly independent of b . The diffusive bias is thus fixed with no free parameter: its size follows from the disk simply being in the overlap regime.

star by its churning $|\Delta L_z|$, recovers the provenance bias (Figure 8). Here H is the spiral’s vertical scale – the simulation’s stand-in for the analytic reach $1/k$ of the $e^{-k|z|}$ fall-off, distinct from the disc scale height $z_0 = h_Z$ – so the simulation’s dimensionless thinness $\alpha \equiv z_0/H$ is the exact counterpart of the analytic $\alpha = kz_0$; the correspondence between the two is only approximate, since the galpy vertical profile is a sech-type form rather than a pure $e^{-k|z|}$, although the ratio z_0/H itself is exact. Migrators are vertically colder, and the bias strengthens with α in agreement with the single-resonance prediction – the log dispersion ratio reaching $\ln \sigma_z^{\text{mig}} / \sigma_z^{\text{all}} \approx -0.06$ (migrators ≈ 6 percent colder) near the Galactic $\alpha \sim 1$. Stacking M overlapping patterns – the pattern count, not the arm number m ; each keeps the same m (and hence k) at a different pattern speed – then deepens the bias monotonically, directly confirming the resonance-overlap mechanism of Section 4.

5. THE PROVENANCE BIAS AND THE $J_R^{\text{max}}-h_Z$ RELATION

Bias magnitude.—In the diffusive regime a star is captured into migration with a probability set by the competition between its radial random energy σ_R^2 and the resonant coupling energy $|\Phi_s|F$ – the spiral amplitude reduced by the vertical form factor of Equation (5). A Boltzmann capture probability $\propto \exp(-\sigma_R^2/|\Phi_s|F)$, nor-

malized to a mid-plane orbit ($F = 1$), gives the effective migration weight

$$W(J_z) = \exp \left[-s \left(\frac{1}{F(J_z, k)} - 1 \right) \right], \quad s \equiv \frac{\sigma_R^2}{|\Phi_s|}, \quad (14)$$

the smooth, three-dimensional form of the K. J. Daniel & R. F. G. Wyse (2015) random-energy capture criterion. Applying the reweighting of Section 3 with this W yields the migrator dispersion ratio $\sigma_z^{\text{mig}} / \sigma_z^{\text{all}}$. At the Milky-Way thickness $\alpha = 0.84$ the predicted bias spans ≈ -20 to -41 percent across the physical range $s \sim 0.8-2$ ($|\Phi_s| \sim 1-3\% V_c^2$), covering the derived diffusive band (≈ -20 to -27 percent; Section 4) with no tuning; the model predicts the bias weakens as the spiral strengthens and strengthens with disk thickness α (Figure 9), matching the strength dependence of D. Mikkola et al. (2020) and the morphology independence of C. Vera-Ciro et al. (2016). As Section 4 makes explicit, the steepness encoded in s is the signature of the diffusive (overlap) regime rather than of clean single-resonance trapping; this effective weight is the phenomenological counterpart of the crossover weight (12) at $S_0 \approx 2-3$.

Structural slope.—Combining the isothermal-sheet scale height $h_Z = \sigma_z^2 / (\pi G \Sigma)$ with the epicyclic action $J_R^{\text{max}} = c_{\text{max}} \sigma_R^2 / \kappa$ gives the slope as the squared radial-to-vertical heating-anisotropy ratio,

$$a = \frac{J_R^{\text{max}}}{h_Z} = c_{\text{max}} \frac{\pi G \Sigma}{\kappa} \left(\frac{\sigma_R}{\sigma_z} \right)^2. \quad (15)$$

For solar-neighborhood values ($\Sigma = 50 M_\odot \text{ pc}^{-2}$, $\kappa = 0.037 \text{ km s}^{-1} \text{ pc}^{-1}$, $\sigma_R = 35$, $\sigma_z = 18 \text{ km s}^{-1}$) and $c_{\text{max}} = 1$, Equation (15) gives $3.6 \times 10^{-2} L_\odot / \text{kpc}$, within a few percent of the measured 3.7×10^{-2} . Equation (15) uses the self-gravitating sech² sheet, whereas P. A. Palicio et al. (2025) fit h_Z as an exponential vertical scale-length; the two scales differ by an order-unity factor, which is absorbed into c_{max} .

Age-running.—Because Σ and κ are structural, the slope’s only population dependence is $(\sigma_R / \sigma_z)^2$. Each channel obeys an age-velocity-dispersion relation $\sigma \propto \text{age}^\beta$, so

$$a(\text{age}) \propto \text{age}^{2(\beta_R - \beta_z)}. \quad (16)$$

With $\beta_z \approx 0.5$ and $\beta_R \approx 0.35$ (M. Aumer et al. 2016; W. Sun et al. 2025), the exponent is -0.3 , so the slope of an 8 Gyr population is ≈ 0.54 times that of a 1 Gyr population: *the relation should flatten with age*. Re-fitting the $J_R^{\text{max}}-h_Z$ relation in mono-age or mono-abundance bins thus measures $\beta_R - \beta_z$ from spiral-arm kinematics; an age-independent slope would instead require $\beta_R = \beta_z$.

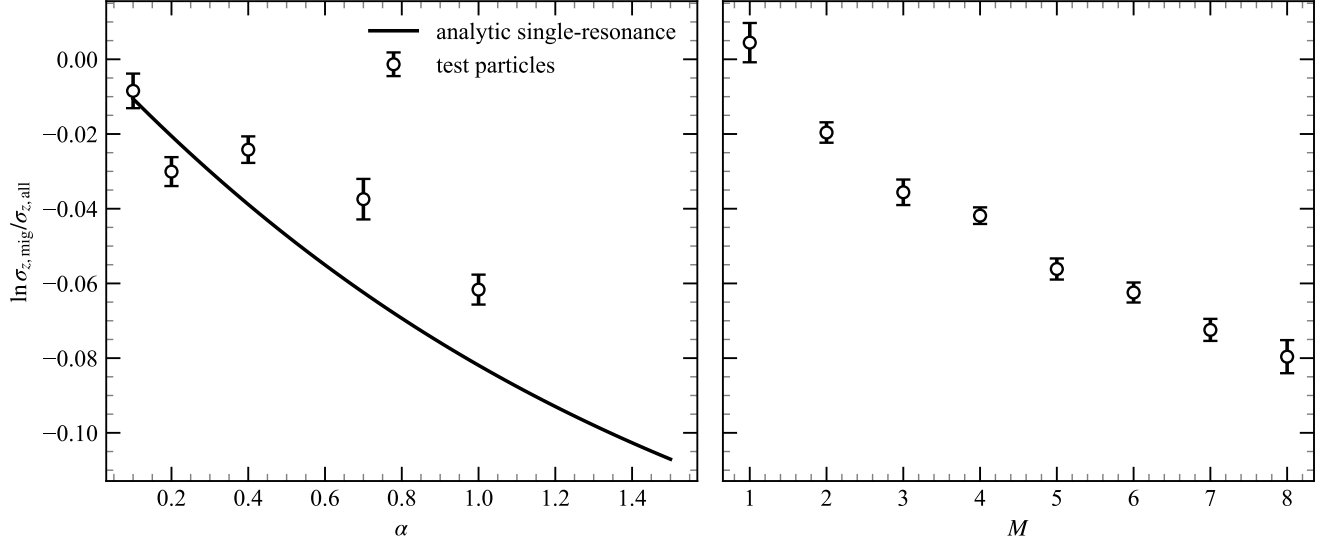


Figure 8. Test-particle validation by direct orbit integration of ~ 3000 particles, with no analytic input (seed-averaged over eight realizations; error bars are the standard error; vertical axis $\ln \sigma_{z,\text{mig}} / \sigma_{z,\text{all}}$, more negative is colder). *Left:* for a single transient spiral the churning-weighted bias deepens with spiral thinness $\alpha = z_0/H$, tracking the analytic single-resonance prediction (curve) and reaching $\ln \sigma_{z,\text{mig}} / \sigma_{z,\text{all}} \approx -0.06$ (migrators ≈ 6 percent colder) near the Galactic $\alpha \sim 1$. *Right:* stacking M overlapping patterns of staggered pattern speed deepens the bias monotonically. Direct simulation thus confirms both halves of the chain – the single-resonance trend and the resonance-overlap mechanism – with no analytic input.

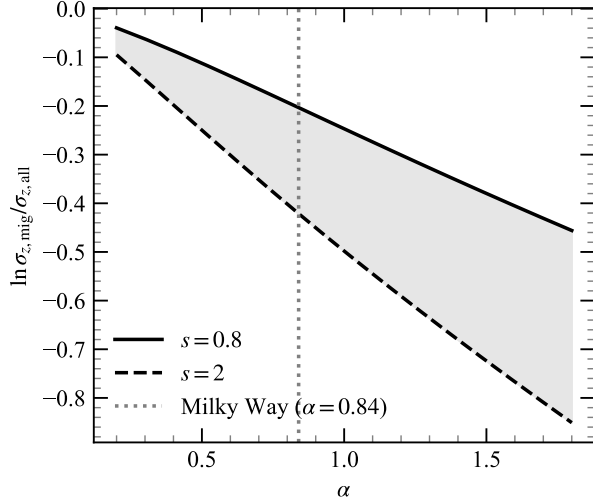


Figure 9. The predicted provenance bias $\ln \sigma_{z,\text{mig}} / \sigma_{z,\text{all}}$ versus disk thickness $\alpha = kh_Z$, from the diffusive capture weight (14), banded over the physical spiral-strength range $s = 0.8\text{--}2$ ($|\Phi_s| \sim 1\text{--}3\% V_c^2$; solid $s = 0.8$, dashed $s = 2$). The bias deepens with thickness, and at the Milky-Way value $\alpha = 0.84$ (dotted) reaches ≈ -20 to -41 percent. Assembled from the form factor alone with no free parameter, the predicted band is consistent with the strong cold-migrator preference seen in the simulations (C. Vera-Ciro et al. 2016; D. Mikkola et al. 2020).

6. DISCUSSION

The provenance bias, the spiral $J_R^{\text{max}}\text{--}h_Z$ relation, and the vertical selectivity of corotation capture are tied together by a single analytic object, the vertical form factor $F(kh_Z)$, which Equation (5) identifies as the exact vertical Fourier coefficient of the spiral in the corotation-resonant Hamiltonian. F is the common thread, not the sole determinant of each magnitude: the bias size additionally requires the migration weight set by the resonance regime (Sections 3–4), while the $J_R^{\text{max}}\text{--}h_Z$ slope follows from the structural anisotropy factor $\pi G \Sigma / \kappa (\sigma_R / \sigma_z)^2$ – the form factor does not enter the slope directly, its role being to select the cold-migrator population that defines J_R^{max} and to share the same disk thickness $\alpha = kh_Z$. Its radial twin is the classical K. J. Daniel & R. F. G. Wyse (2015) in-plane selectivity; the $e^{-k|z|}$ reduction is standard (J. Binney & S. Tremaine 2008); and the bias was found in simulations (C. Vera-Ciro et al. 2016; D. Mikkola et al. 2020). What was missing is the vertical extension that turns the verbal mechanism into Equations (5)–(8) and connects it to P. A. Palicio et al. (2025). The form factor should not be conflated with the finite-thickness reduction factors of classical density-wave theory (A. Toomre 1964), which weaken a self-gravitating disk’s *collective* response to a density wave: F is instead the orbit-average of an externally imposed spiral felt by a *single* star at fixed J_z

– a test-particle coupling, not a modification of the self-gravitating response.

A complementary first-principles link between migration and vertical structure already exists: R. Roškar et al. (2013) show, from vertical-action conservation, that an outward migrator’s scale height grows as $h_{z,f}/h_{z,i} = \exp(\Delta R/2R_d)$. That law governs the *fate* of a star once it has migrated; the form factor governs the *selectivity* of capture – which stars migrate at all. The two are distinct and complementary, and only the latter produces the provenance bias.

The two dynamical regimes the form factor distinguishes – single-resonance trapping ($W \propto F^{1/2}$) and resonance-overlap diffusion (W steeper) – are the two ends of one axis. The diffusive end is the inhomogeneous Balescu–Lenard description of secular evolution in tightly wound disks (J.-B. Fouvry et al. 2015), and the linear-response perturbed distribution function of G. Monari et al. (2016) is its complement; a full treatment would compute $W(J_z)$ as the phase-space trapped fraction across the overlap transition rather than in its two limits, which is left to future work.

Several caveats bound the calculation. The finite-thickness softening of the $e^{-k|z|}$ kernel (Section 2, Equa-

tion (2)) is an order-unity correction at the Galactic $\alpha \sim 1$, carried here for an exponential spiral profile; a self-consistent profile would refine it. The overlap parameter (11) is sensitive to the bar–spiral pattern-speed separation; the well-separated fiducial adopted here is conservative, and the Milky Way’s recurrent transient spirals add resonances that only increase the overlap. Finally, the provenance-bias magnitude is not yet robustly measured: C. Vera-Ciro et al. (2016) establish the cold-migrator preference qualitatively, not as a quantitative dispersion offset. The conclusions here rest on the regime ordering – single-resonance trapping floors near -7 percent, the diffusive regime predicts -20 to -27 percent – and on the qualitative cold-migrator preference, rather than on a precise value. The age-running of the slope (16) is, to my knowledge, an untested and clean prediction for current Gaia–APOGEE–LAMOST data.

DATA AVAILABILITY

The open-source engine that derives every number and figure in this paper, with a full test suite, is available at <https://github.com/caverac/jr-hz-relation>; a step-by-step walkthrough of the derivation is hosted at <https://caverac.github.io/jr-hz-relation/>.

APPENDIX

A. THE SOFTENED (FINITE-THICKNESS) FORM FACTOR

A spiral arm is overdense disk material of finite vertical extent rather than a razor-thin sheet. Model its vertical mass profile as a normalized exponential $g(z) = (\beta/2) e^{-\beta|z|}$ with $\beta \equiv 1/h_s$ ($\int g dz = 1$), and keep the in-plane pattern $\Sigma_0 e^{ikx}$, so $\rho(x, z) = \Sigma_0 e^{ikx} g(z)$. Writing $\Phi = e^{ikx} \psi(z)$ reduces Poisson’s equation to $\psi'' - k^2 \psi = 4\pi G \Sigma_0 g(z)$; the Green’s function of $\partial_z^2 - k^2$ is the razor-thin kernel itself, $\mathcal{G}(z) = -e^{-k|z|}/2k$, so the potential is that kernel convolved with the source,

$$\psi(z) \propto (e^{-k|\cdot|} * g)(z) = \frac{\beta(k e^{-\beta|z|} - \beta e^{-k|z|})}{k^2 - \beta^2}. \quad (\text{A1})$$

Normalizing to $K(0) = 1$ gives the softened kernel $K(z) = [\beta e^{-k|z|} - k e^{-\beta|z|}]/(\beta - k)$ used in Section 2. The orbit average is linear in the z -dependence, and each exponential is itself a form factor at its own wavenumber, $\langle e^{-k|z|} \rangle = F(J_z, \alpha)$ and $\langle e^{-\beta|z|} \rangle = F(J_z, 1/h)$ with $h \equiv h_s/z_0$; hence $\langle K \rangle$ is Equation (2). It recovers the razor-thin F as $h \rightarrow 0$ and tends to unity as $h \rightarrow \infty$, when the star never escapes the arm.

B. THE EXACT TRAPPED FRACTION

The pendulum (6) has separatrix $p_{\text{sep}}(\theta_r) = 2\sqrt{\Psi/g} |\sin(\theta_r/2)|$; since $\Psi \propto F$ (5), $p_{\text{sep}} = p_{\text{max}} \sqrt{F} \sin(\theta_r/2)$ with $p_{\text{max}} = 2\sqrt{\Psi_0/g}$ the $F=1$ half-width. For a parent population Gaussian in the slow action, $N(p) \propto e^{-p^2/2\sigma_p^2}$, the fraction trapped at fixed θ_r (those with $|p| < p_{\text{sep}}$) is $\text{erf}(p_{\text{sep}}/\sqrt{2}\sigma_p)$. Averaging over θ_r and folding to $u = \theta_r/2 \in (0, \pi)$, with the single island-width parameter $\kappa \equiv p_{\text{max}}(F=1)/\sigma_p \propto \sqrt{|\Phi_s|}$, yields the exact trapped weight, Equation (9). The narrow-island limit ($\kappa \rightarrow 0$, $\text{erf } x \rightarrow 2x/\sqrt{\pi}$) recovers the island-counting law $W \propto \sqrt{F}$ (8), while the wide-island limit ($\kappa \rightarrow \infty$, $\text{erf} \rightarrow 1$) gives $W \rightarrow 1$ and washes out the selectivity.

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